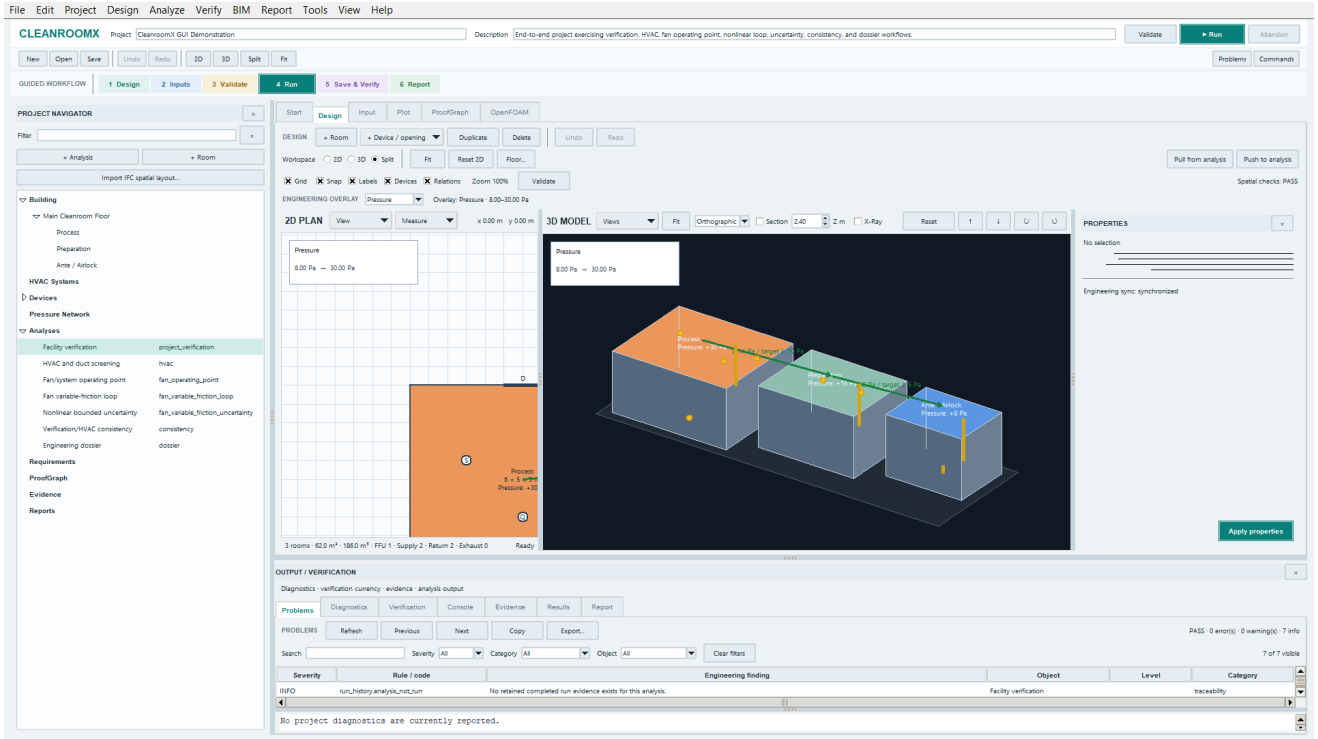
OPERATOR EDITION

CLEANROOMX

Illustrated GUI Manual

Real application screenshots • light and dark modes • project workflow • engineering evidence boundaries



Companion to CX-UM-001 | Screenshot set: 17 Oct 2026 | Scope: GUI demonstration, not CFD validation

About this illustrated manual

A navigational companion for the existing controlled CleanroomX engineering handbook.

SOURCE OF TRUTH

All figures originate from PNG files uploaded from the running CleanroomX Windows GUI. The original files were reviewed for pixel dimensions and SHA-256 duplicates. This is documentation of the interface, not independent verification of simulation results.

WHAT THIS ADDS

Visual identification of project navigation, 2D/3D editing, focus mode, analysis inputs, output panels and theme controls. Commands and safe operating sequences complement, not replace, the full manual.

SCIENTIFIC BOUNDARY

The example fan operating point is a calculated numerical example, not OpenFOAM CFD. The supplied files named OpenFOAM or ProofGraph currently show the Results panel, not those workspaces. Their functionality is not demonstrated by these images.

SCREENSHOT AUDIT

18 uploaded PNG files comprise 13 unique image hashes. Some images have a 1920 × 1055 pixel client capture, while others are 1920 × 1080. No images were enlarged or altered to conceal these differences.

Important: “PASS” in a screen may concern spatial checks, application diagnostics or integrity checks only. It must not be read as evidence of independent CFD validation.

From project launch to engineering evidence

A safe and repeatable first-use workflow using the bundled demonstration project.

1

Open CleanroomX

From a repository checkout on Windows, run .\start-cleanroomx.ps1 --demo, or use cleanroomx-gui --demo after installation.

2

Inspect the workspace

Open the Start tab and locate project controls, the navigator, and the Design tab. The demo contains three illustrative rooms.

3

Review geometry

Choose 2D, 3D or Split. Inspect the pressure overlay and check spatial validation status; do not assume geometry has been certified.

4

Select a numerical study

Choose Fan/system operating point under Analyses. Inspect editable inputs before execution.

5

Run and review

Execute the example and use Plot / Results for graphs and values. Record input provenance, execution status, findings, and version.

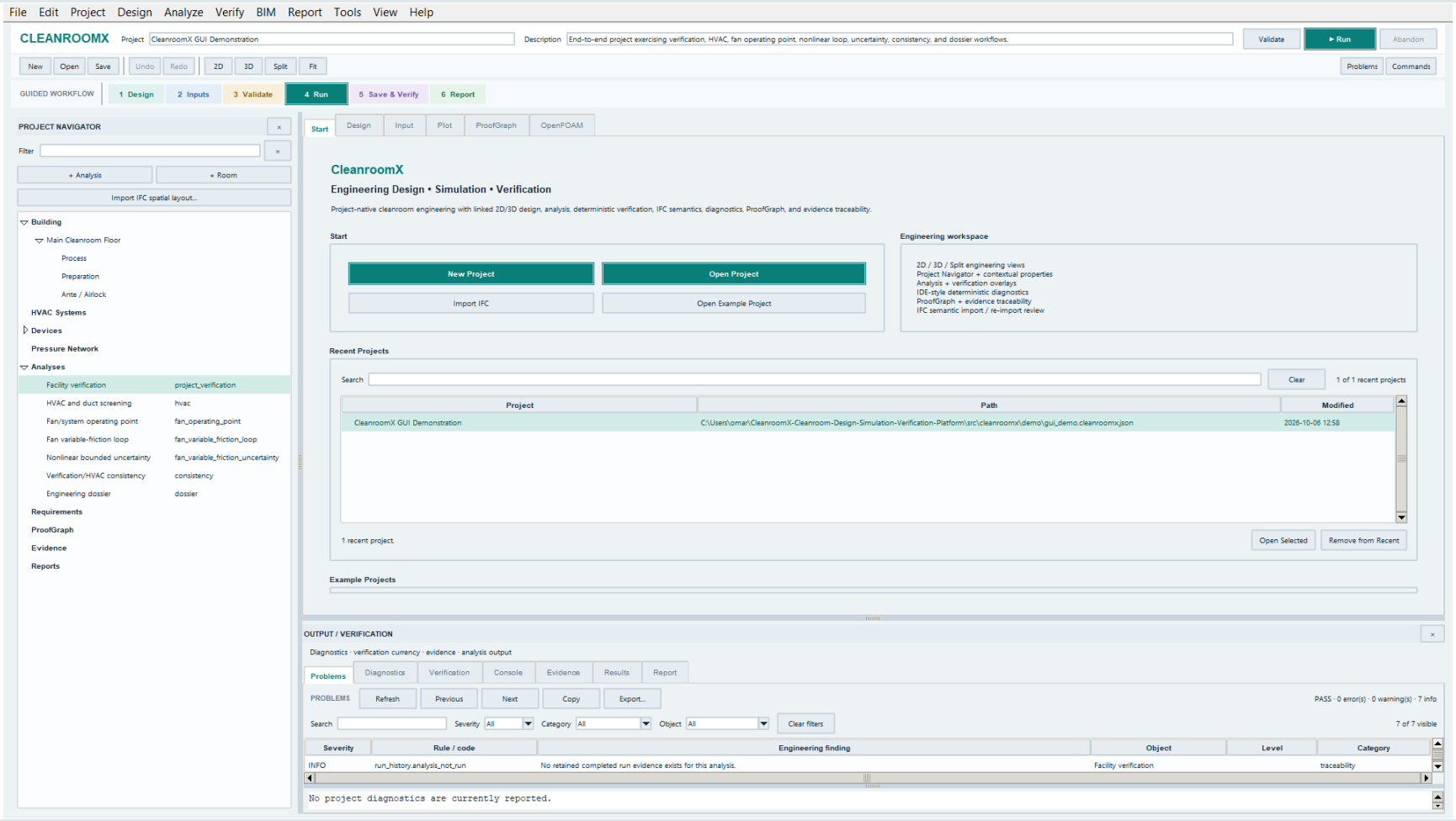
6

Keep evidence honest

OpenFOAM solver output, scientific validation, and ProofGraph provenance require their own auditable inputs and review.

Start center | light mode

Open a project, inspect recent projects, or load the bundled demo.



WHAT YOU SEE

Visible project navigator on the left, a start workspace with open/new actions, and an output/verification panel.

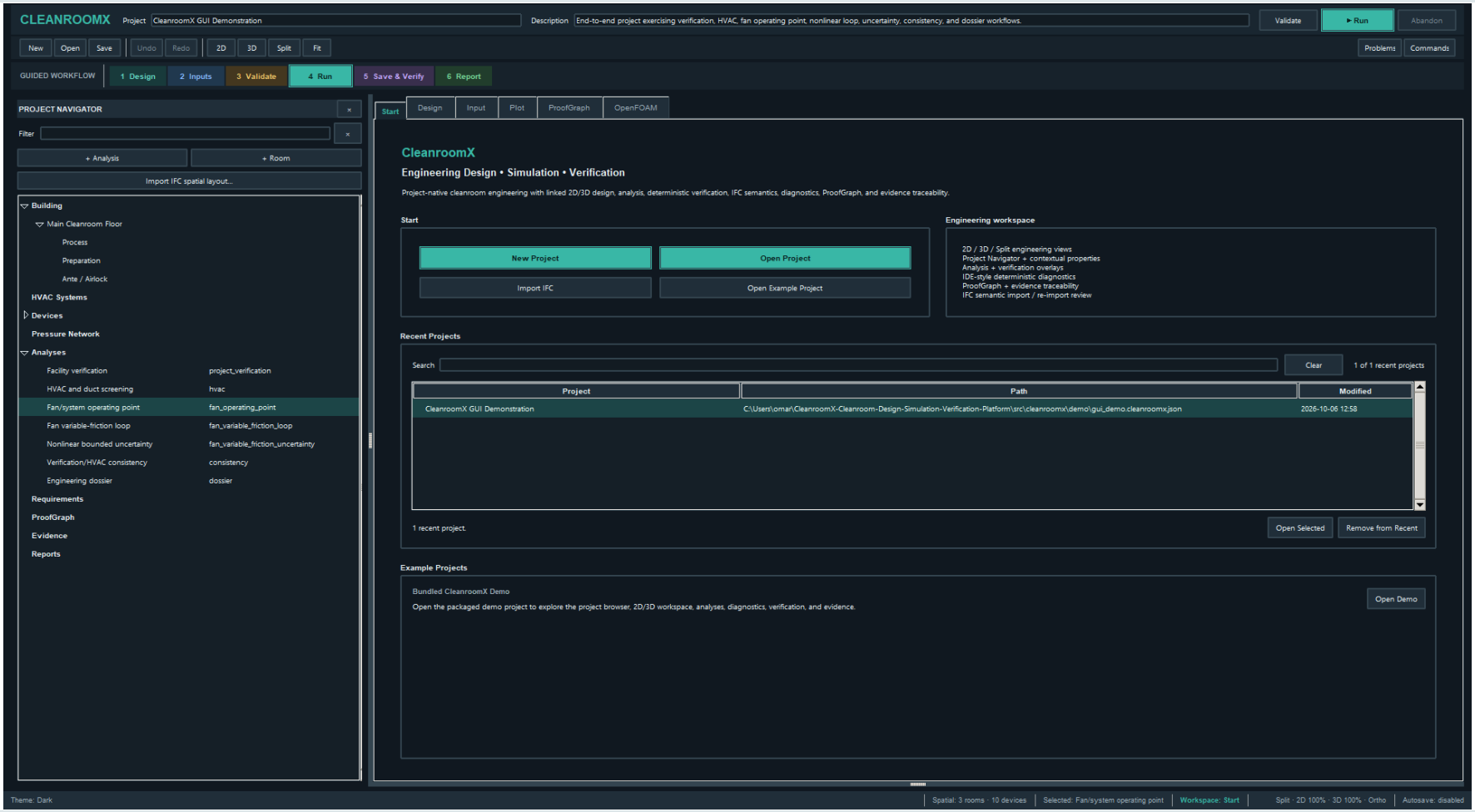
Source capture: 1920 × 1080 (full window).

OPERATOR ACTION

Select an existing project or the bundled demo; inspect diagnostics before engineering work.

Start center | dark mode

A darker display palette for the same project controls.



WHAT YOU SEE

Start tab and recent-project access remain available with dark widget colors.

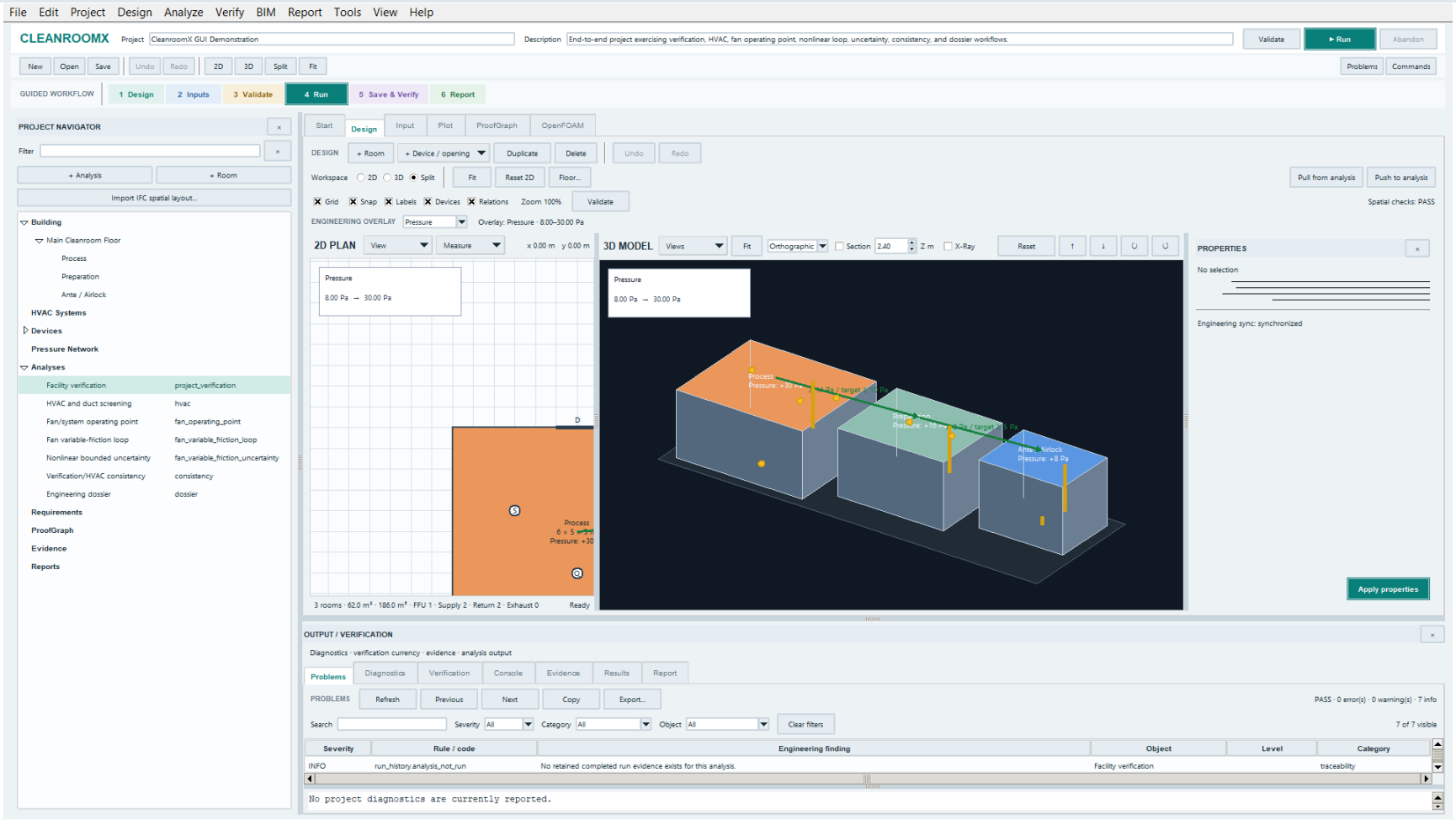
Source capture: 1920 × 1055 (client-area screenshot).

OPERATOR ACTION

Use the theme setting to improve readability; color alone does not change engineering inputs.

Design workspace | contextual layout

2D and 3D displayed together, with navigation and results areas.



WHAT YOU SEE

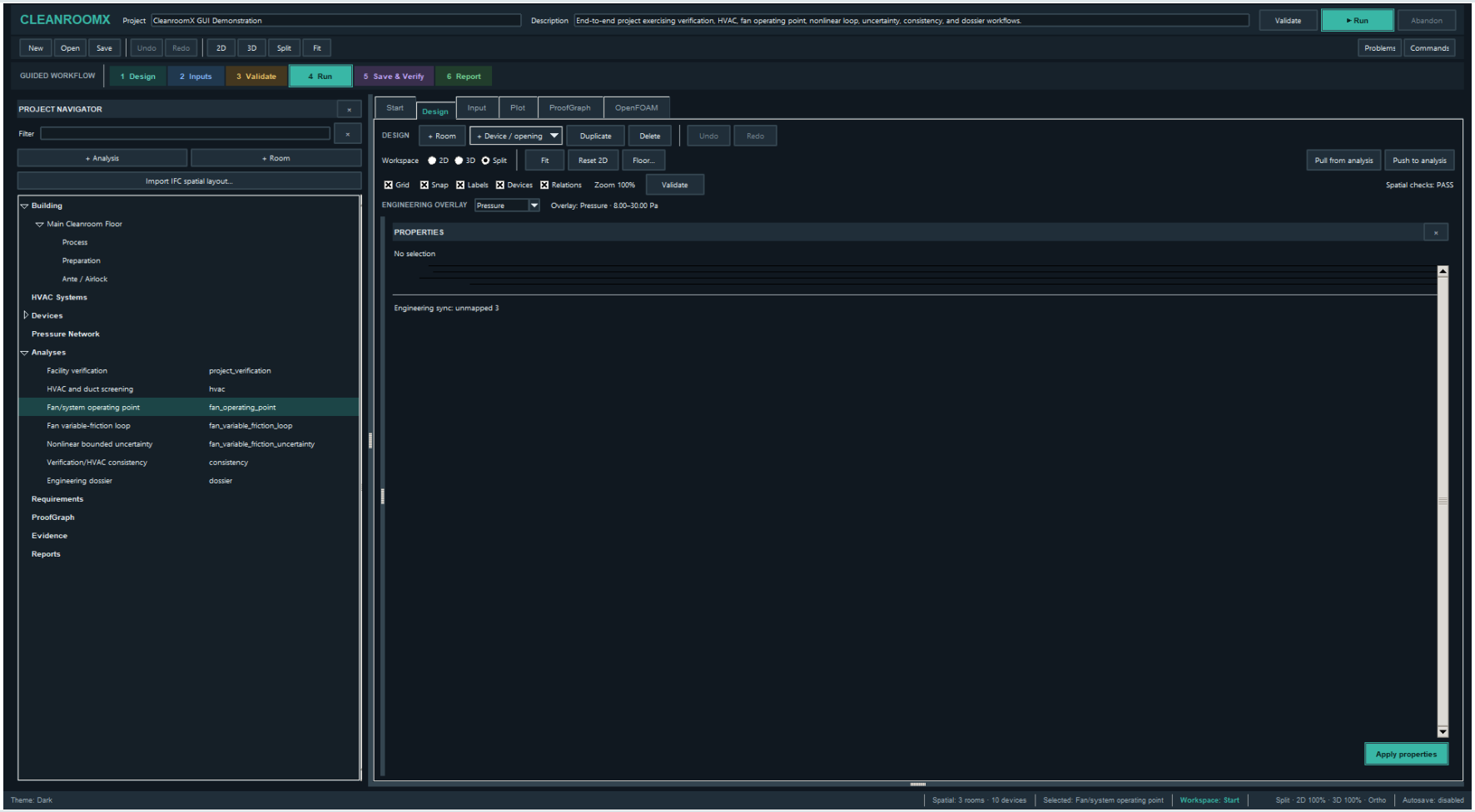
Project Navigator at left, 2D and 3D workspaces in the center, properties at right, and verification at bottom.

OPERATOR ACTION

Begin here to assess data context, spatial model status, and the scope of unsolved analyses.

Design workspace | dark contextual layout

Properties and navigation remain visible in dark mode.



WHAT YOU SEE

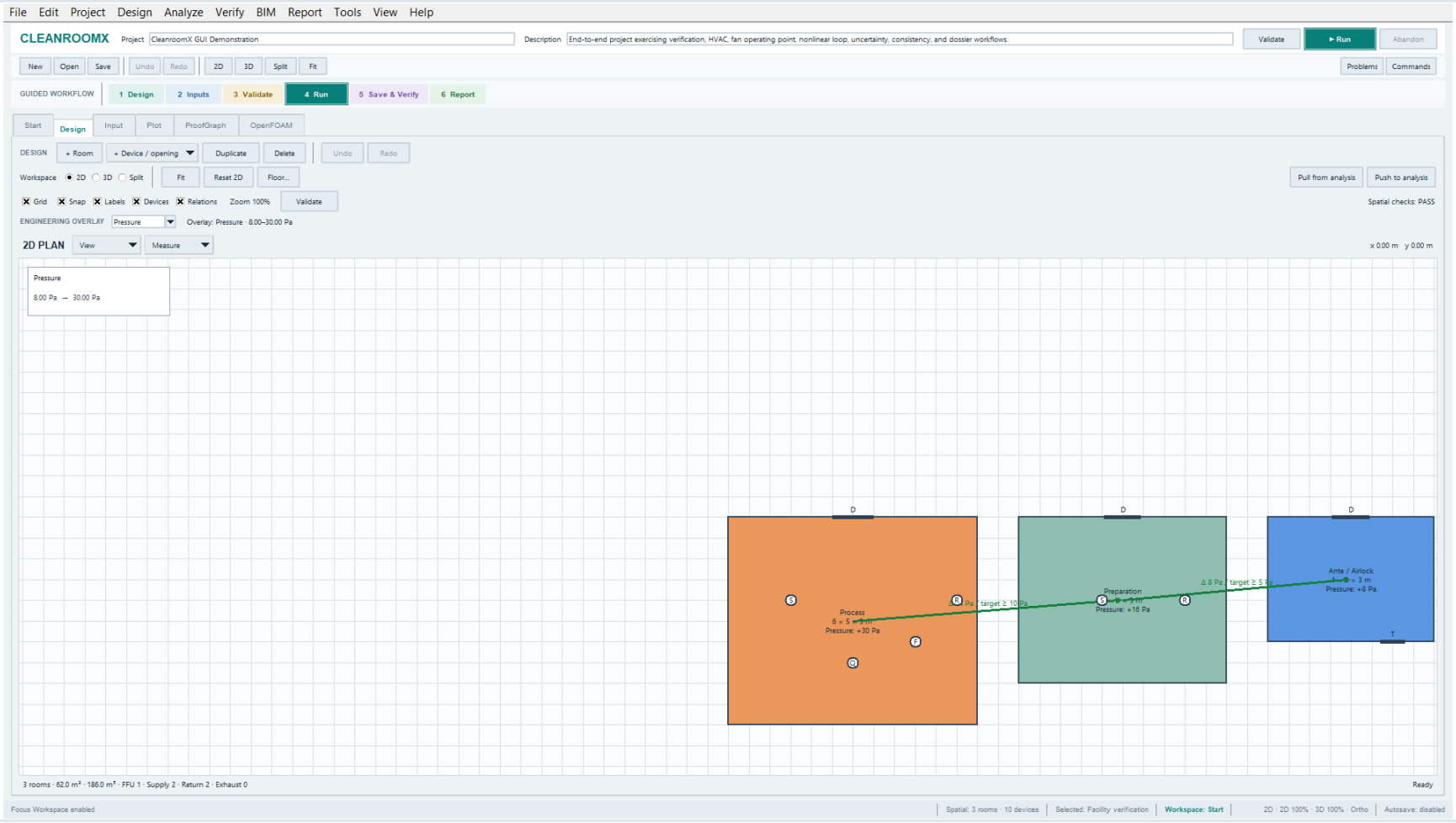
Dark theme design/inspector layout. The uploaded screenshot shows an unmapped engineering-sync status for the current selection.

OPERATOR ACTION

Treat “unmapped” as a coordination gap; verify explicit spatial-to-engineering mapping.

2D plan | light mode

Room footprints and a pressure-cascade overlay.



WHAT YOU SEE

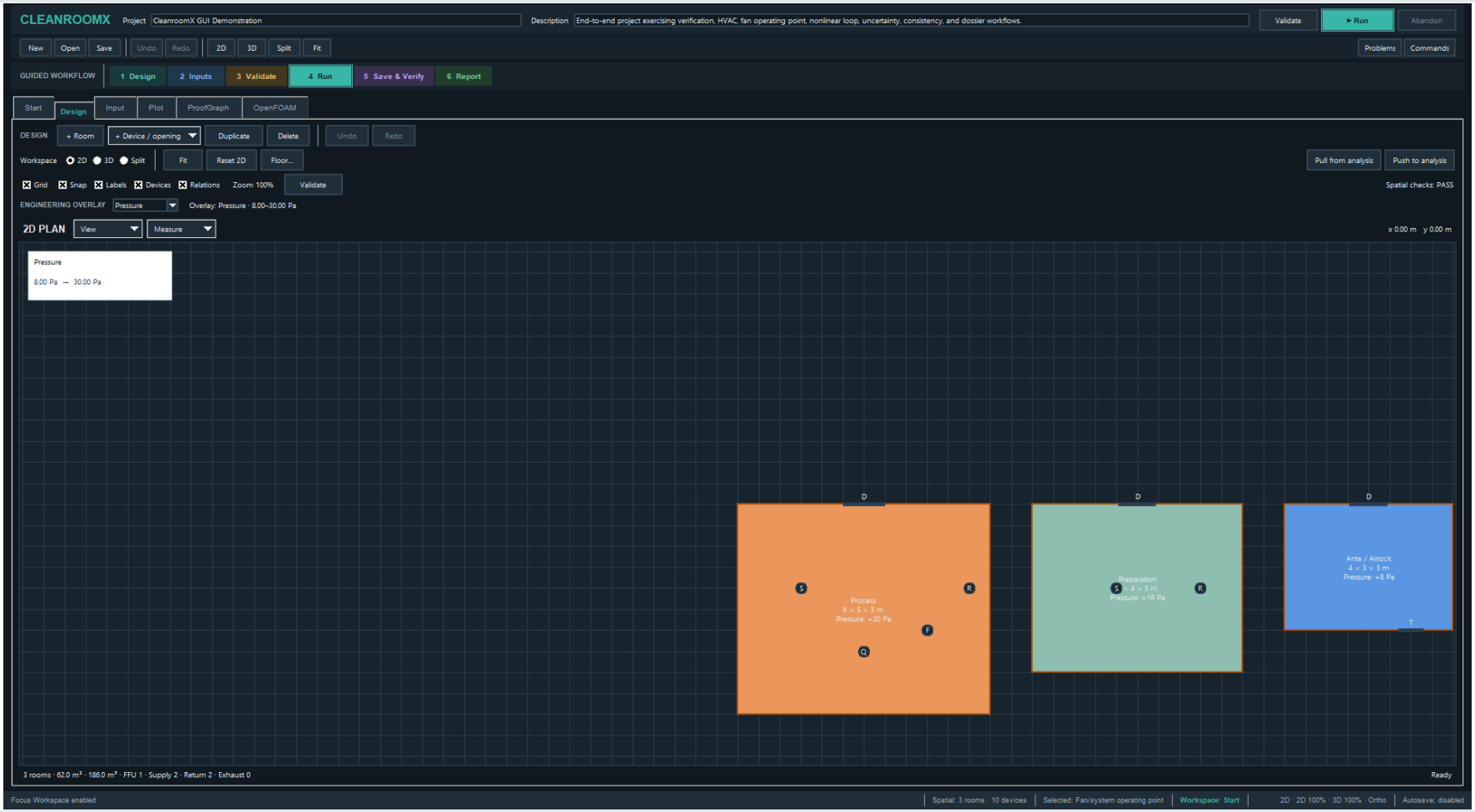
Three demo rooms are arranged on a gridded floor plan, with arrows between successive spaces.

OPERATOR ACTION

Inspect labels, room dimensions, doors and the engineering overlay; confirm actual values from project inputs.

2D plan | dark mode

An enlarged viewport for fine spatial review.



WHAT YOU SEE

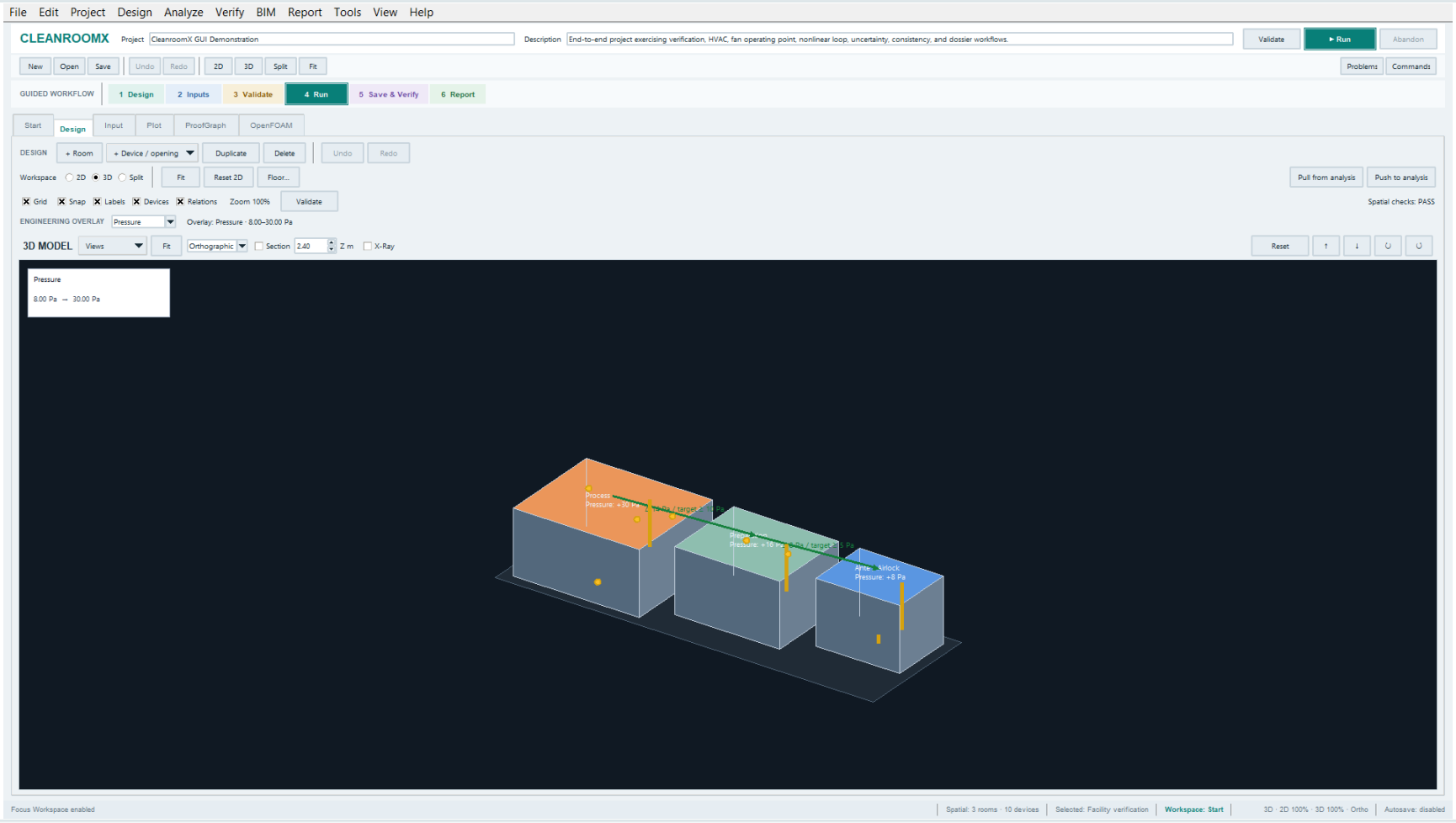
The 2D-only view preserves the plan geometry, pressure labels and room connection marks.

OPERATOR ACTION

Use Fit or Reset 2D to inspect the full layout and check if small labels remain readable.

3D spatial model | light mode

A three-room illustrative cleanroom model.



WHAT YOU SEE

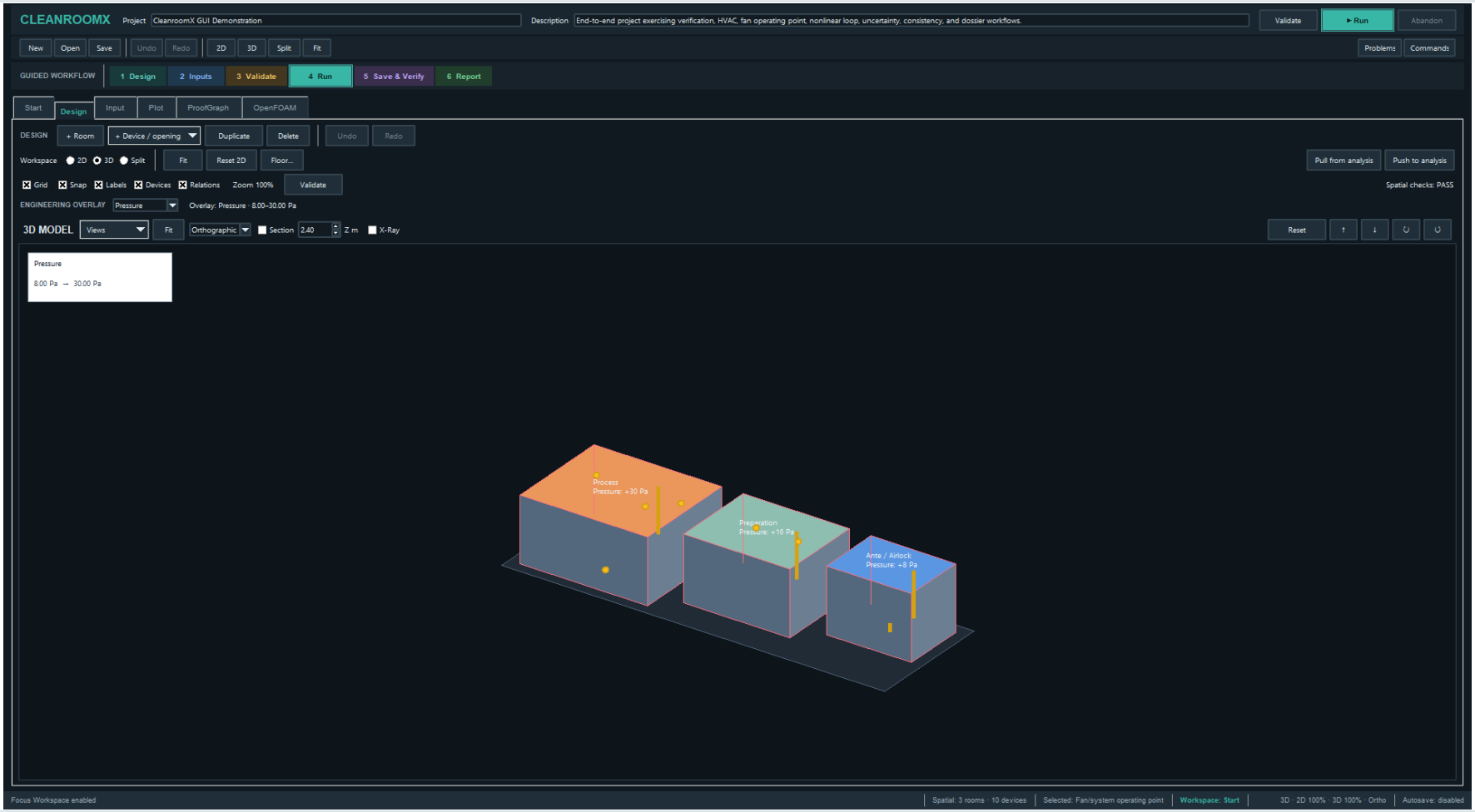
The viewport contains three raised room solids, color-coded tops, pressure labels and device/connection glyphs.

OPERATOR ACTION

Rotate and zoom the 3D view to check spatial relationships and equipment positions.

3D spatial model | dark mode

The same example geometry with a dark viewport.



WHAT YOU SEE

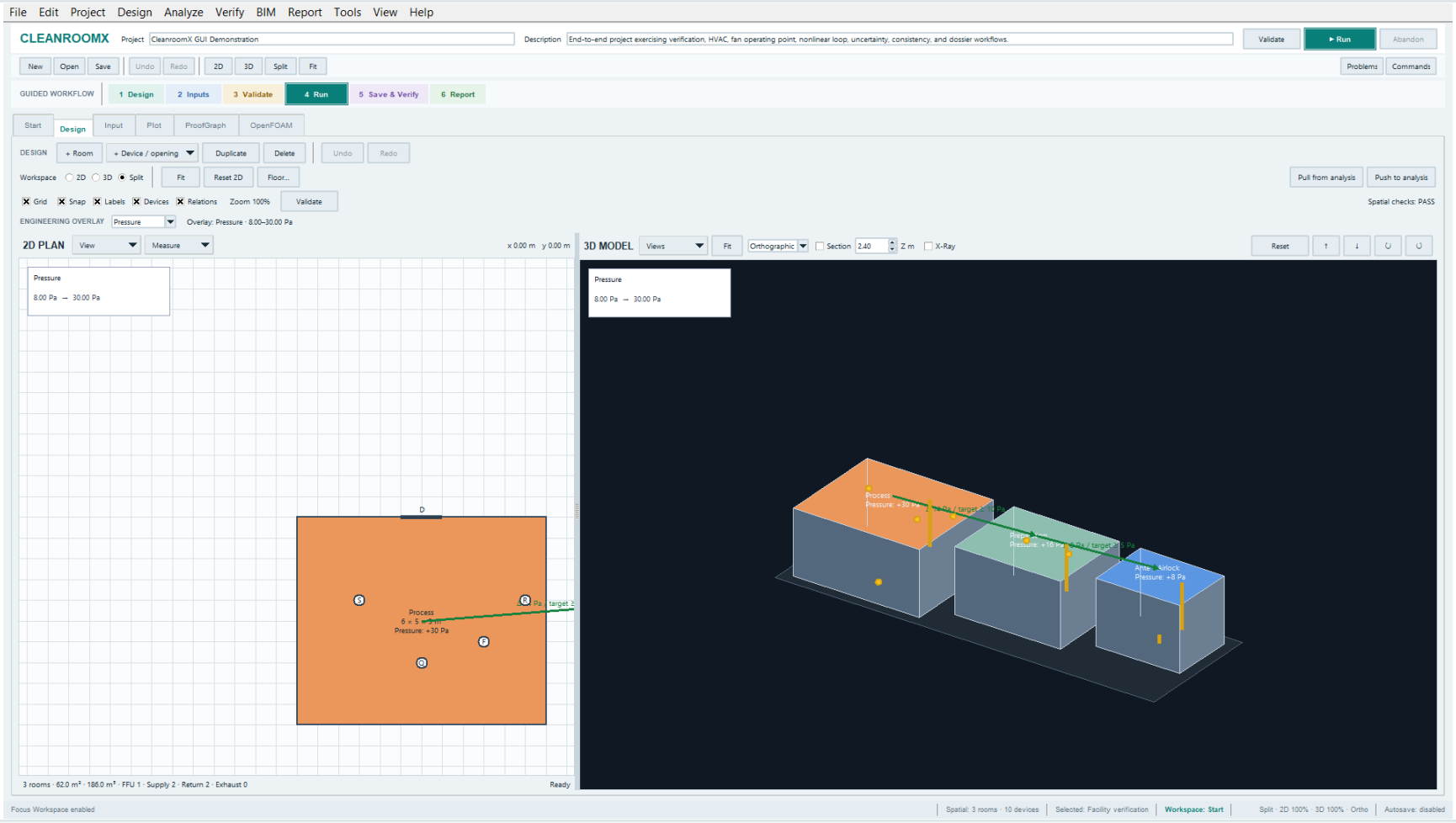
Dark scene demonstrates the application renderer and alternative visual contrast.

OPERATOR ACTION

Do not infer airflow uniformity or turbulence from a 3D room-solid view. This is design visualization.

Split view | enlarged workspace

Synchronize 2D floor plan and 3D room representation.



WHAT YOU SEE

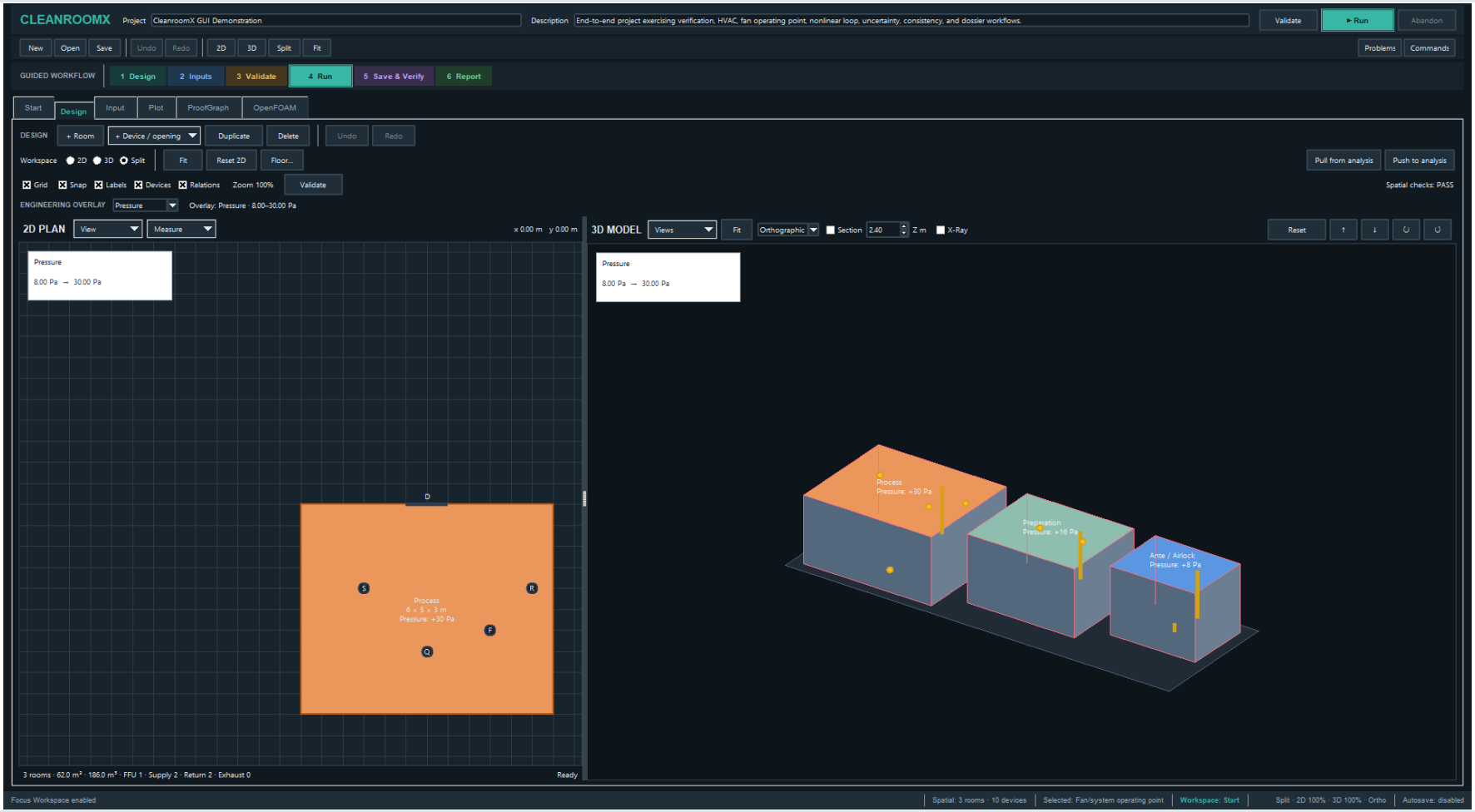
Split mode places a scaled plan on the left and the modeled spaces on the right.

OPERATOR ACTION

Cross-check room adjacency and placement across both views before pushing changes to analyses.

Split view | dark mode

Reduced visual clutter with navigation panels hidden.



WHAT YOU SEE

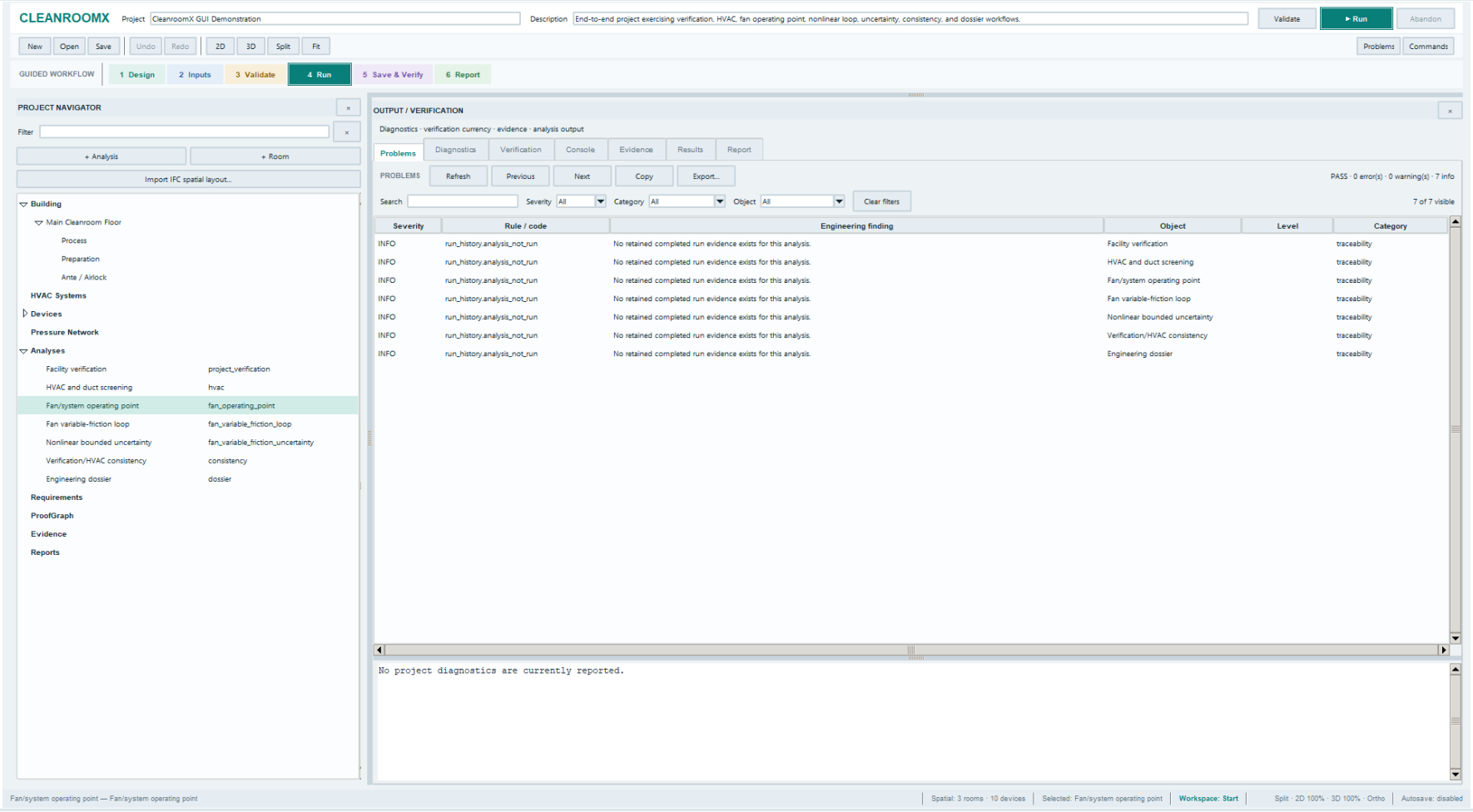
The focused split workspace provides more space for the plan and 3D model.

OPERATOR ACTION

Use Focus Workspace for presentation; restore panels for input editing and traceability review.

Analysis selection and inputs | light

The analysis navigator exposes the fan study, but the visible pane is Results.



WHAT YOU SEE

The left navigation highlights Fan/system operating point. The main content shows solved numerical JSON rather than the input editor.

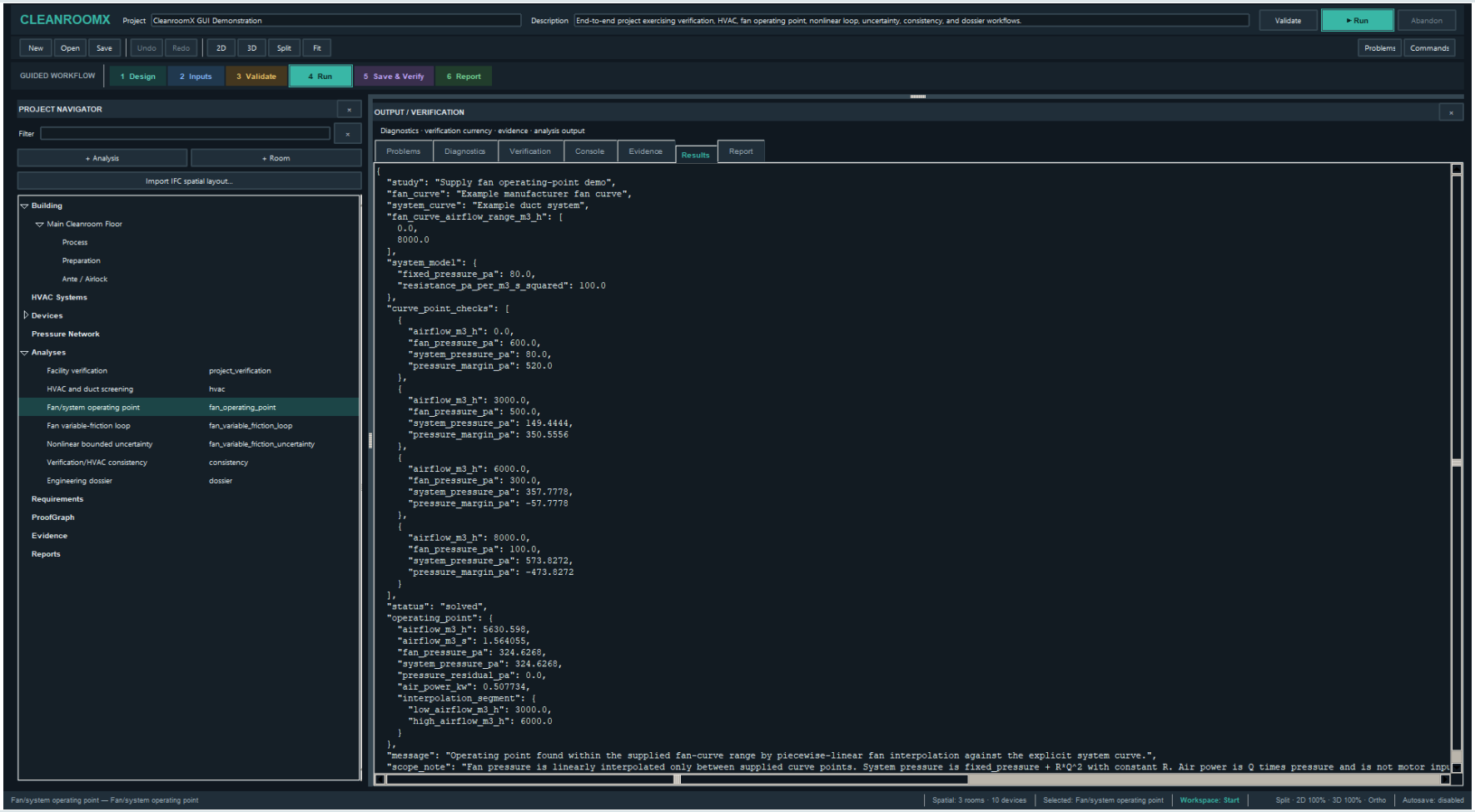
Caption deliberately distinguishes the selected study from the visible output panel.

OPERATOR ACTION

Open the Input tab to review editable parameters before running a new case.

Analysis selection and inputs | dark

The selected fan analysis alongside its result JSON.



WHAT YOU SEE

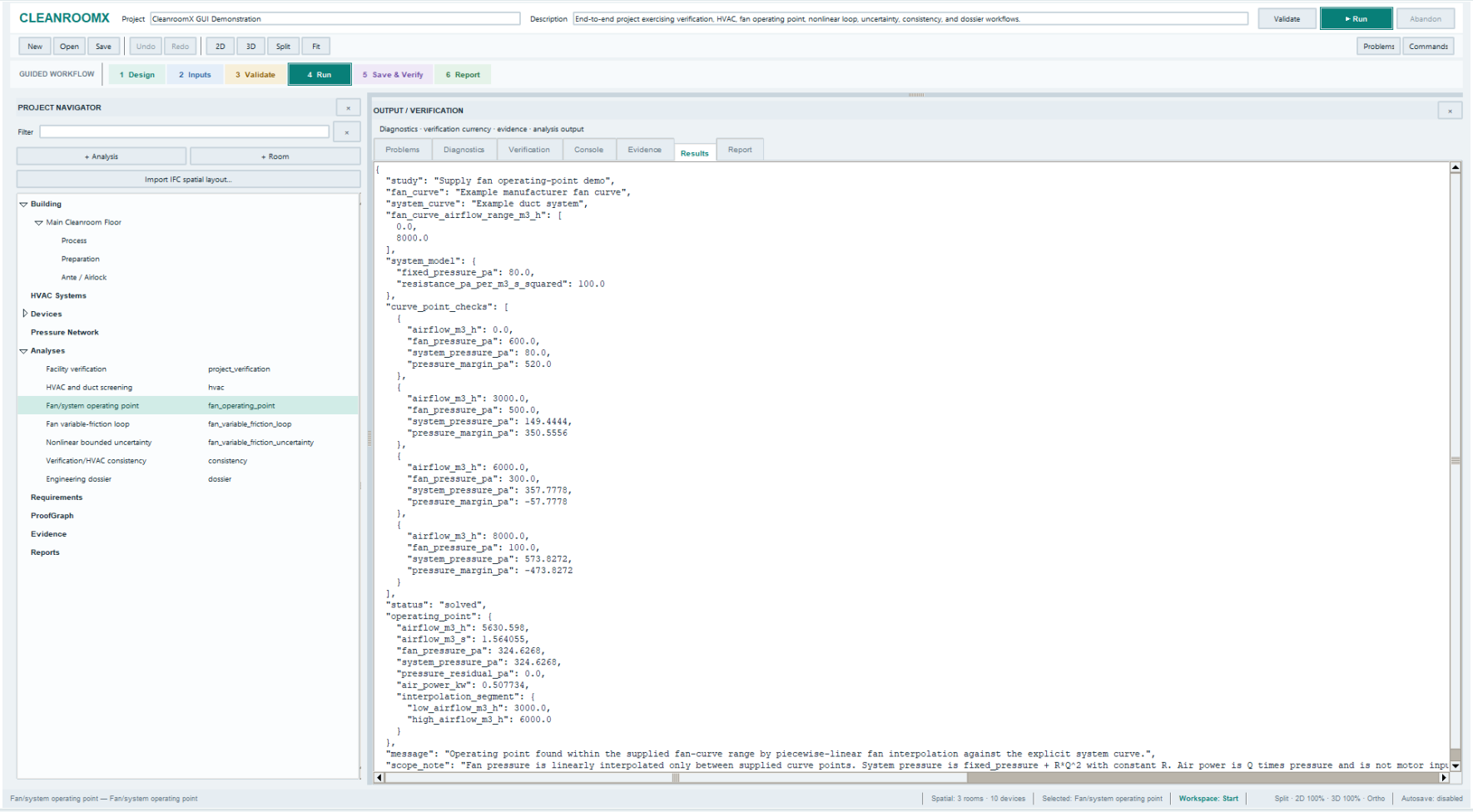
The screenshot actually displays the output/verification Results view, not the Input tab.

OPERATOR ACTION

Use the selected study name to verify which run the displayed numerical output belongs to.

Computed fan results | numerical output

A backend-calculated operating-point example, not CFD.



WHAT YOU SEE

The image shows JSON numerical results and “solved” status; it does not show a plotted curve.

This source screenshot duplicates several mislabeled OpenFOAM and ProofGraph files.

OPERATOR ACTION

Switch to the Plot tab to inspect the fan and system curves and their intersection.

Capture limitations | OpenFOAM and ProofGraph

This release of the illustrated manual does not invent missing interface screenshots.

What the uploaded filenames suggest

gui-openfoam-workbench.png and gui-openfoam-proofgraph-diagnostic.png imply views of the OpenFOAM and ProofGraph workspaces.

What the actual pixels show

Both show the same fan operating-point Results pane. They cannot substantiate a screenshot-based explanation of OpenFOAM execution, imported CFD field visualizations or ProofGraph diagnostics.

To complete these sections with real captures

1. Select the OpenFOAM tab and ensure the OpenFOAM controls are visibly mapped.
2. Capture a distinct, complete window image; retain the unmodified PNG and its hash.
3. Select ProofGraph with the relevant evidence loaded; capture a separate view.
4. Label imported/demo data accurately and never treat a synthetic fixture as validated CFD.

Audit finding: 18 filenames -> 13 unique image SHA-256 values; five files duplicate other captures.

Using the GUI responsibly

Different panes answer different engineering questions.

Design

What is modeled? Use 2D/3D room geometry, devices, doors and pressure overlays.

Input

What assumptions are supplied? Inspect analysis selection, units, boundary conditions and configuration.

Run / Plot

What did the backend compute? Check completed runs, numerical output, plots and diagnostics.

ProofGraph

What evidence supports a claim? Trace artifacts, hashes, mappings and scientific validation boundaries.

Review sequence: geometry -> input contract -> solver or numerical run -> diagnostics -> evidence bundle -> engineering sign-off.

Source screenshot register

SHA-256 shown as a short identifier for review; full digests stored in companion JSON.

01	gui-start-center.png	1920 x 1080	04b3986313258c
02	gui-start-center-dark.png	1920 x 1055	7cd63526bc19d4
03	gui-design-split-context.png	1920 x 1080	77bc3e7c5d27d4
04	gui-design-split-context-dark.png	1920 x 1055	b4a688c436dd0f
05	gui-design-2d.png	1920 x 1080	1d0bc1781a6fc7
06	gui-design-2d-dark.png	1920 x 1055	52d1ba7efcfaac
07	gui-design-3d.png	1920 x 1080	27c45bacee4df6
08	gui-design-3d-dark.png	1920 x 1055	1ca4e039891ed6
09	gui-design-split.png	1920 x 1080	5c2af3c0677e9e
10	gui-design-split-dark.png	1920 x 1055	84c6f2b693126c
11	gui-analysis-inputs.png	1920 x 1055	5270d990a8b5da
12	gui-analysis-inputs-dark.png	1920 x 1055	21644a587e025b
13	gui-computed-fan-results.png	1920 x 1055	3f87079b8268e8

Five duplicate PNGs are preserved in the companion package and documented in its JSON audit.

Publication checklist

A GUI illustration is not a scientific-verification receipt.

1

Trace the image

Confirm screenshot filename, dimensions, content hash, selected tab and data provenance.

2

Check the message

Captions must correspond to what the screenshot actually shows. A filename does not prove content.

3

Keep the numerical boundary

Computed fan curves are not CFD. Synthetic synchronization is not evidence of solver convergence.

4

Review outstanding gaps

Obtain real OpenFOAM and ProofGraph workspace images before advertising those sections as illustrated.

5

Publish with provenance

Store original PNGs, a manifest with hashes and the PDF. Record software revision and validation status separately.